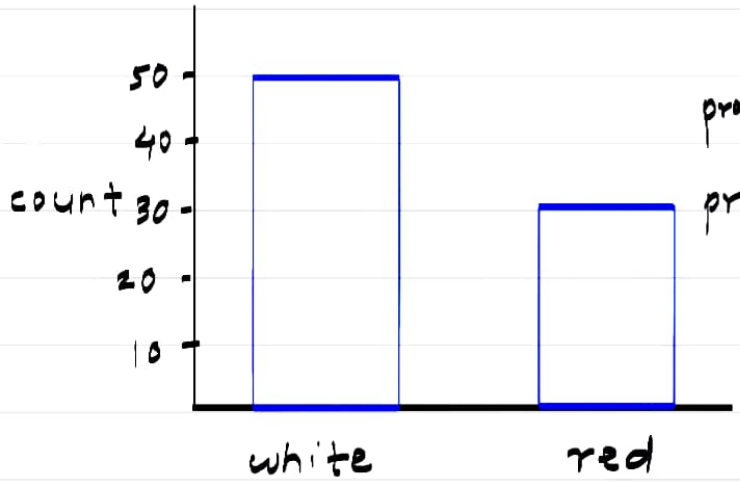
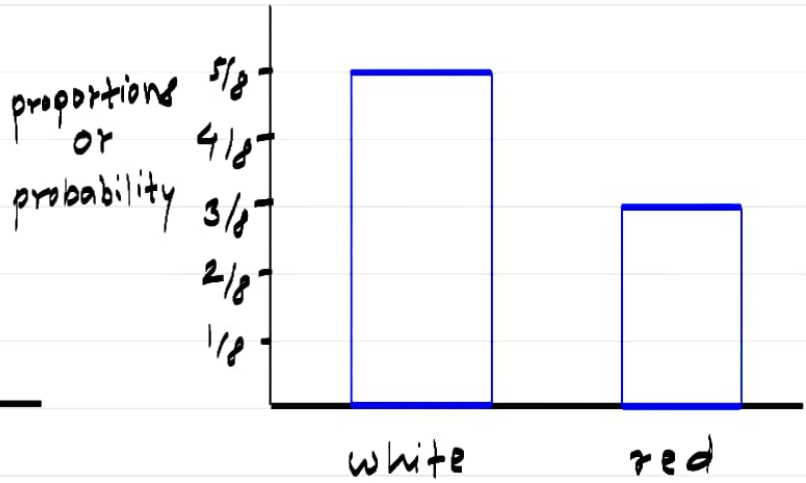
Advanced statistics

20-01-26

Categorical data

bag of balls - 50 white + 30 red





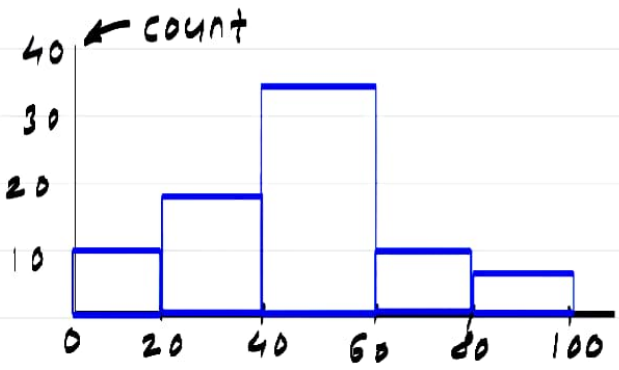
proportion of white $= \frac{50}{80} = \frac{5}{8}$

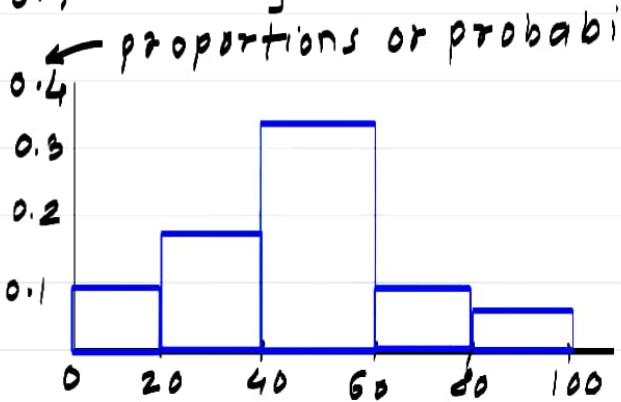
proportion of red $= \frac{30}{80} = \frac{3}{8}$

we can see that
probability distribution is same as
data distribution

Continuous data

Age = [10, 20, 11.7, 15.3, 67, 81, ...] total values = 100

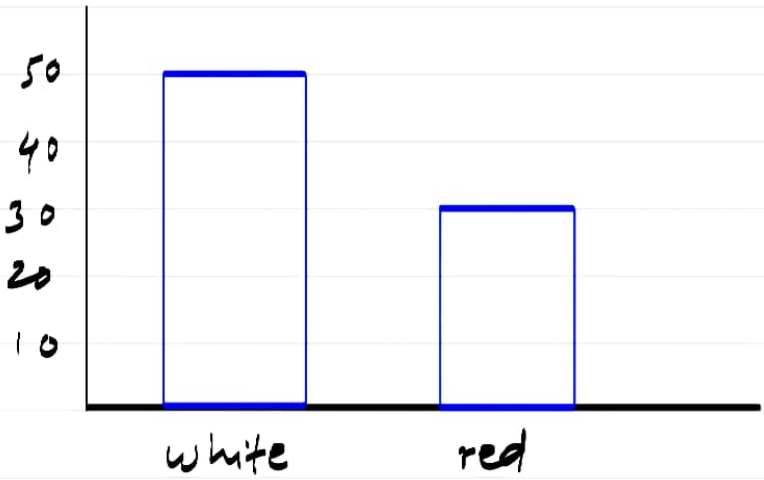




- Every underlying data distribution has a probability distribution associated to it

- Thats why mathematical expression / function of a probability distribution can be also used as mathematical representation for data distribution

categorical data
50 white + 30 red balls



- As this data has 2 categories, it can be associated to a random experiment having 2 possible outcomes
- when a random experiment has 2 possible outcomes, one of them is called success(1) and the other one as failure (0)

lets say white - success (1) red - failure (0)

$p(\text{success}) = p(\text{white}) = \frac{50}{80} = 5/8$

$p(1) = p = 5/8$

$p(\text{failure}) = p(\text{red}) = 30/80 = 3/8$

$p(0) = 1 - p = 1 - 5/8 = 3/8$

$$\left.\begin{aligned} p(1) &= p = 5/8 \\ p(0) &= 1-p = 3/8 \end{aligned}\right\} \quad \begin{aligned} f(x) &= p^x (1-p)^{1-x} \\ &\text{where } p = 5/8 \end{aligned}$$

input $\longrightarrow$ $\boxed{\text{function}}$ $\longrightarrow$ output

sample functions

$f(x) = x^2+1$

$f(1) = 1^2+1 = 2$

$f(3) = 3^2+1 = 10$

$f(-2) = -2^2+1 = 5$

probability function for given data

$$f(x) = p^x (1-p)^{1-x} \quad , \quad p = 5/8$$

$f(1) = p^1 (1-p)^{1-1} = p(1-p)^0 = p \times 1 = p$

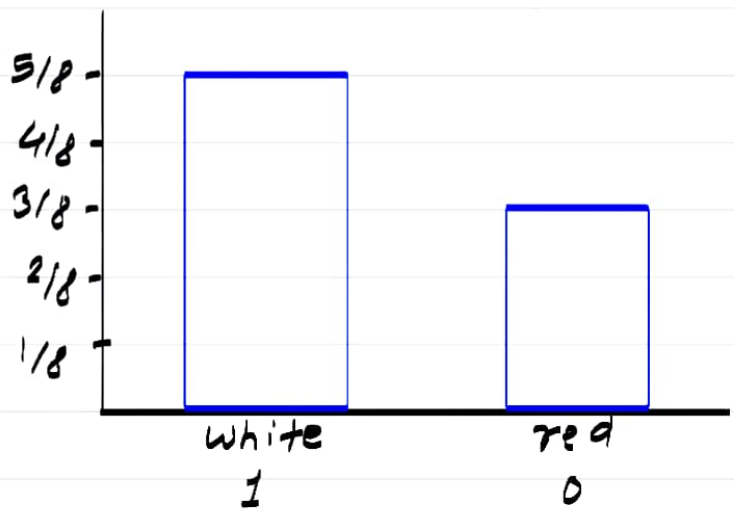
$f(1) = p = 5/8$

$f(0) = p^0 (1-p)^{1-0} = 1 \times (1-p)$

$f(0) = 1-p = 1 - 5/8 = 3/8$

$f(x) = p^x (1-p)^{1-x}$

where $p = 5/8$

$f(1) = 5/8$

$f(0) = 3/8$